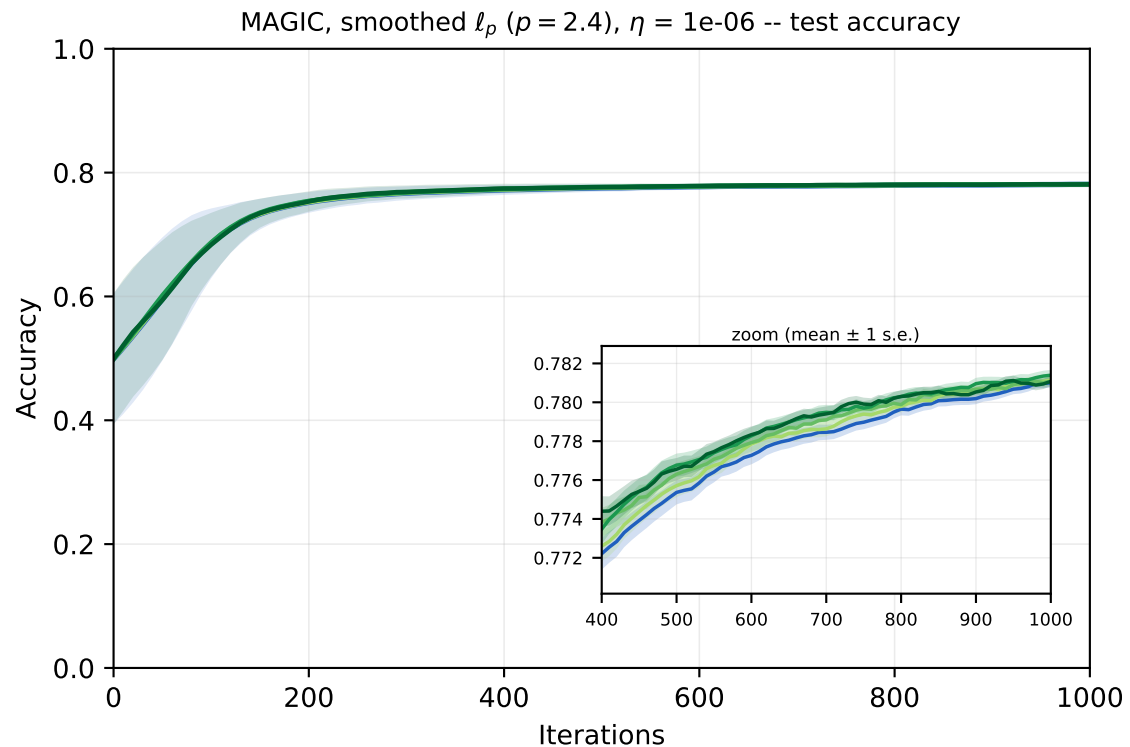
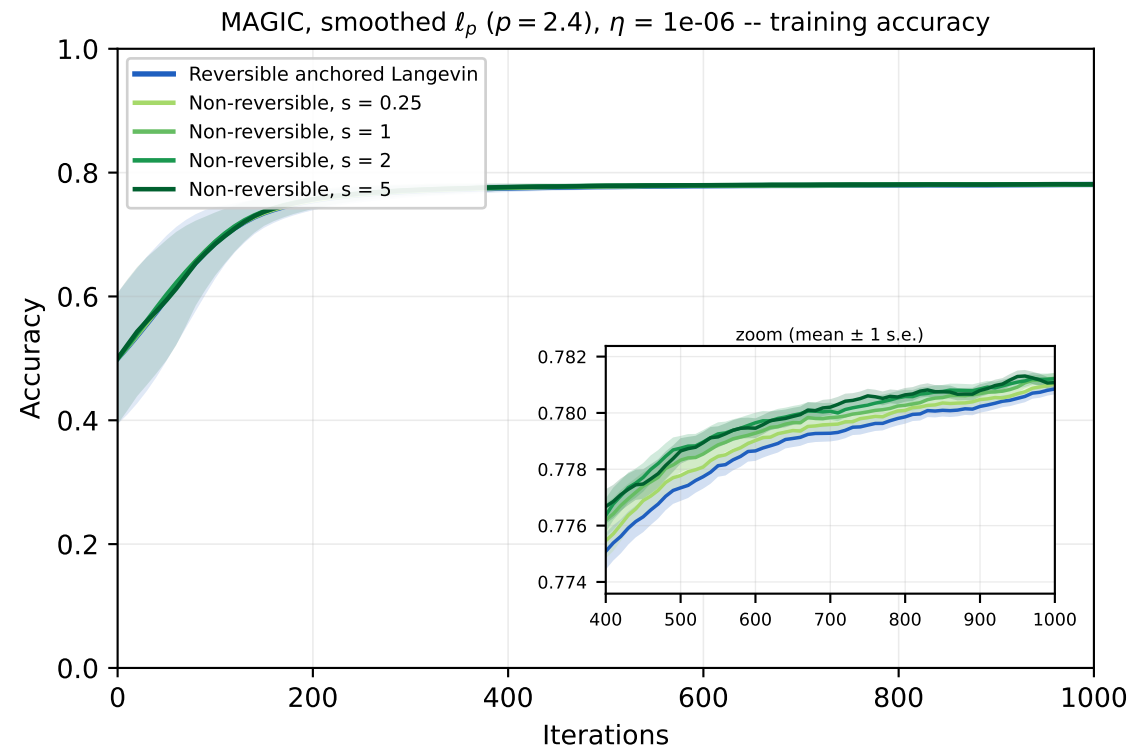


# magic\_lp: accuracy vs block strength at $\eta = 1e-06$



MAGIC -- smoothed  $\ell_p$   $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{p/2} \leq \text{Lambda}$ ,  $p = 2.4$ ,  $\text{eps} = 0.2$ ,  $\text{Lambda} = 4$ .  $d = 10$  (intercept  $w_0 + 9$  features).  $U = \sum \text{softplus}(-y x \cdot w + w_0^2 / (2 \sigma^2)) + \text{lambda} \sum_{j \geq 1} |w_j|$ ;  $U_0$  replaces  $|w_j|$  by  $\sqrt{w_j^2 + \text{delta}^2}$ ;  $a = \exp(U - U_0)$ . EXACT gradient.  $\sigma = 10$ ,  $\text{lambda}_{\text{lasso}} = 152 = 0.01 n_{\text{train}}$ ,  $\text{delta} = 0.000506$  ( $a$  in  $[0.50, 1]$ ). Step  $\eta = 1e-06$ .  $R = 100$  independent replicates on one fixed stratified 80/20 split ( $n_{\text{train}} = 15216$ ,  $n_{\text{test}} = 3804$ ). Bands: mean  $\pm 1$  SD (ddof = 1) for the reversible arm and for  $s = 5$  -- repeat-run variability at a fixed split, not a confidence or credible interval. The whole  $\eta \times s$  grid is reported in `results/grid/grid.csv`; nothing here was selected.